

## MOSSTRAND II, Norway

WMO: 014500

Lat: 59.839N

Lon: 8.179E

Elev: 3205

StdP: 13.07

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |            | Humidification DP/MCDB and HR |            |             |             |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|------------|-------------------------------|------------|-------------|-------------|------------|------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |            | 99.6%                         |            |             | 99%         |            |            | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%        | DP                            | HR         | MCDB        | DP          | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-3.8 | (c)<br>0.2 | (d)<br>-9.7                   | (e)<br>3.7 | (f)<br>-2.3 | (g)<br>-5.6 | (h)<br>4.6 | (i)<br>1.1 | (j)<br>42.8           | (k)<br>28.7 | (l)<br>38.0 | (m)<br>27.7 | (n)<br>2.4            | (o)<br>320 | (p)<br>0.800 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.3                   | 70.5            | 55.2 | 66.9 | 53.6 | 63.2 | 52.2 | 57.1                | 66.6 | 55.3 | 63.4 | 53.7 | 60.6 | 8.1                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 53.7                            | 69.1 | 58.6 | 51.9 | 64.7 | 57.6 | 50.2 | 60.6 | 55.9 | 25.8          | 67.1 | 24.6 | 63.2 | 23.5 | 60.7 | 63.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 32.9          | (b) 27.5 | (c) 23.3 | (e) -8.4                   | (f) 75.1 | (g) 4.5            | (h) 3.4  | (i) -11.6                                          | (j) 77.6  | (k) -14.3  | (l) 79.6  | (m) -16.8  | (n) 81.5  | (o) -20.0  | (p) 83.9  |    |    |
| (4) 32.9          | (5) 27.5 | (6) 23.3 | (7) -8.4                   | (8) 75.1 | (9) 4.5            | (10) 3.4 | (11) -11.6                                         | (12) 77.6 | (13) -14.3 | (14) 79.6 | (15) -16.8 | (16) 81.5 | (17) -20.0 | (18) 83.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 35.3   | 19.4  | 20.5  | 25.8  | 32.1  | 41.0  | 48.8  | 53.3  | 50.4  | 45.0  | 35.8  | 27.7  | 22.7  |
|                                                                        | DBStd       | 13.62  | 8.86  | 8.98  | 8.02  | 6.49  | 7.35  | 5.27  | 5.63  | 3.92  | 4.83  | 5.50  | 7.05  | 9.12  |
|                                                                        | HDD50       | 5626   | 949   | 826   | 749   | 536   | 295   | 84    | 26    | 42    | 164   | 442   | 668   | 845   |
|                                                                        | HDD65       | 10840  | 1414  | 1246  | 1214  | 986   | 744   | 486   | 363   | 452   | 601   | 906   | 1118  | 1310  |
|                                                                        | CDD50       | 261    | 0     | 0     | 0     | 0     | 16    | 48    | 128   | 55    | 13    | 1     | 0     | 0     |
|                                                                        | CDD65       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 24     | 0     | 0     | 0     | 0     | 0     | 2     | 22    | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 8.9    | 9.9   | 9.4   | 10.1  | 8.0   | 7.4   | 8.5   | 7.9   | 8.1   | 9.3   | 9.2   | 9.4   | 10.0  |
|                                                                        | PrecAvg     | 31.1   | 3.0   | 1.9   | 1.7   | 1.5   | 2.0   | 2.7   | 3.5   | 3.6   | 3.1   | 3.0   | 2.7   | 2.5   |
| (15) Precipitation                                                     | PrecMax     | 42.7   | 6.5   | 5.0   | 4.4   | 3.2   | 5.4   | 6.7   | 7.3   | 6.5   | 5.9   | 7.8   | 7.0   | 5.8   |
|                                                                        | PrecMin     | 22.6   | 0.4   | 0.1   | 0.1   | 0.4   | 0.1   | 0.4   | 1.1   | 0.3   | 1.0   | 0.5   | 0.7   | 0.2   |
|                                                                        | PrecStd     | 4.8    | 1.5   | 1.2   | 1.0   | 0.9   | 1.2   | 1.5   | 1.5   | 1.5   | 1.2   | 1.7   | 1.5   | 1.4   |
|                                                                        | PrecStd     | 4.8    | 1.5   | 1.2   | 1.0   | 0.9   | 1.2   | 1.5   | 1.5   | 1.5   | 1.2   | 1.7   | 1.5   | 1.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 38.7  | 41.1  | 48.4  | 53.4  | 68.7  | 71.9  | 77.2  | 68.5  | 64.5  | 54.1  | 44.8  | 41.4  |
|                                                                        |             | MCWB   | 35.5  | 35.9  | 39.3  | 42.5  | 52.7  | 53.9  | 56.8  | 55.4  | 52.9  | 47.3  | 40.6  | 37.3  |
|                                                                        | 2%          | DB     | 36.1  | 37.7  | 42.8  | 48.9  | 64.1  | 68.3  | 72.6  | 64.4  | 59.5  | 50.1  | 41.2  | 39.1  |
|                                                                        |             | MCWB   | 33.6  | 33.6  | 36.5  | 40.1  | 51.2  | 51.7  | 56.6  | 54.0  | 50.5  | 44.0  | 38.5  | 36.2  |
|                                                                        | 5%          | DB     | 33.6  | 35.3  | 39.1  | 44.8  | 58.4  | 64.2  | 69.1  | 60.9  | 55.3  | 46.4  | 38.9  | 36.8  |
|                                                                        |             | MCWB   | 31.5  | 32.2  | 34.4  | 37.8  | 47.4  | 50.3  | 55.6  | 51.9  | 48.7  | 42.2  | 36.7  | 34.4  |
|                                                                        | 10%         | DB     | 31.0  | 32.5  | 36.2  | 41.7  | 53.6  | 59.8  | 64.6  | 57.8  | 52.5  | 43.7  | 36.7  | 33.8  |
|                                                                        |             | MCWB   | 29.5  | 30.1  | 32.3  | 36.4  | 44.6  | 48.5  | 53.9  | 50.8  | 47.5  | 40.3  | 34.8  | 32.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 35.9  | 37.2  | 40.3  | 44.4  | 54.8  | 55.8  | 60.6  | 57.2  | 54.9  | 51.0  | 41.7  | 39.7  |
|                                                                        |             | MCDB   | 37.9  | 40.1  | 46.9  | 51.5  | 65.8  | 67.7  | 70.0  | 64.2  | 60.6  | 53.9  | 43.9  | 40.9  |
|                                                                        | 2%          | WB     | 33.8  | 34.6  | 37.0  | 41.2  | 51.6  | 53.6  | 58.3  | 55.3  | 51.8  | 45.3  | 39.2  | 36.6  |
|                                                                        |             | MCDB   | 35.6  | 37.0  | 41.6  | 46.7  | 61.4  | 63.5  | 68.3  | 60.9  | 56.8  | 47.8  | 41.1  | 38.3  |
|                                                                        | 5%          | WB     | 31.8  | 32.4  | 34.7  | 38.7  | 48.8  | 51.9  | 56.6  | 53.9  | 50.0  | 43.1  | 37.0  | 34.4  |
|                                                                        |             | MCDB   | 33.1  | 34.6  | 38.5  | 44.1  | 56.6  | 60.8  | 66.2  | 59.5  | 54.1  | 45.8  | 38.6  | 36.1  |
|                                                                        | 10%         | WB     | 29.5  | 30.0  | 32.6  | 36.8  | 46.0  | 50.0  | 54.9  | 52.0  | 48.2  | 40.6  | 34.7  | 31.9  |
|                                                                        |             | MCDB   | 30.9  | 32.6  | 35.9  | 41.4  | 52.4  | 58.2  | 63.1  | 56.4  | 52.2  | 43.2  | 36.6  | 33.4  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.7   | 11.4  | 13.2  | 14.7  | 15.9  | 16.6  | 15.3  | 13.2  | 11.9  | 10.1  | 8.2   | 9.3   |
|                                                                        |             | MCDBR  | 10.9  | 12.7  | 15.0  | 18.9  | 23.6  | 24.4  | 23.4  | 18.1  | 17.0  | 15.2  | 10.1  | 9.7   |
|                                                                        |             | MCWBR  | 9.5   | 10.4  | 10.0  | 11.3  | 12.8  | 11.6  | 9.9   | 9.3   | 9.8   | 10.3  | 8.5   | 9.1   |
|                                                                        | 5% WB       | MCDBR  | 10.1  | 11.0  | 13.7  | 16.8  | 21.2  | 20.1  | 19.5  | 15.2  | 15.1  | 11.7  | 9.2   | 9.7   |
|                                                                        |             | MCWBR  | 9.1   | 9.7   | 9.4   | 10.3  | 12.1  | 10.6  | 9.2   | 9.0   | 9.4   | 9.5   | 8.2   | 9.3   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.208 | 0.244 | 0.272 | 0.307 | 0.312 | 0.307 | 0.322 | 0.316 | 0.289 | 0.273 | 0.223 | 0.181 |
|                                                                        | taud        |        | 2.202 | 2.300 | 2.285 | 2.300 | 2.460 | 2.532 | 2.495 | 2.501 | 2.571 | 2.515 | 2.275 | 2.078 |
|                                                                        | Ebn at Noon |        | 200   | 252   | 277   | 285   | 292   | 294   | 286   | 280   | 272   | 241   | 190   | 161   |
|                                                                        | Edh at Noon |        | 12    | 20    | 28    | 34    | 31    | 30    | 30    | 28    | 22    | 16    | 11    | 9     |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 102   | 306   | 764   | 1282  | 1519  | 1673  | 1491  | 1176  | 765   | 361   | 132   | 66    |
|                                                                        | RadStd      |        | 21    | 69    | 92    | 198   | 209   | 162   | 195   | 96    | 99    | 66    | 31    | 7     |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (61 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page